CAP Theorem - CP Databases (Consistency + Partition Tolerance)

1. What is a CP System?

CP means:

```None
C = Consistency

P = Partition Tolerance
```

System guarantees:

- Latest correct data
- Handles network partitions

But may sacrifice:

```None
Availability
```

Some requests may fail, wait, or be rejected during partition.

2. Simple Meaning

If cluster splits into two groups:

```None
Node A,B | Node C,D
```

CP system chooses correctness over always responding.

It may disable minority nodes or stop writes until quorum forms.

3. Why CP Is Chosen

Used when wrong data is worse than temporary unavailability.

Examples:

Banking balances
Payment ledger
Inventory reservation
Ticket booking
Metadata/config systems
Identity systems

4. Example Scenario

Account balance = ₹10,000

Network partition occurs.

If two sides process withdrawals independently, balance may become wrong.

CP system may instead do:

```None
Reject writes until majority available
```

This preserves correctness.

5. How CP Systems Work

Common mechanisms:

A. Leader Election

One node becomes writer.

B. Majority Quorum

Write accepted only if majority confirms.

C. Replica Sync

Followers replicate leader state.

D. Failover

If leader dies, elect new leader.

6. MongoDB Clarification

MongoDB can behave as a CP-oriented system depending on configuration.

Replica set model:

```None
Primary node handles writes

Secondary nodes replicate
```

If primary fails:

- Election chooses new primary
- During election, writes may pause

That reflects consistency-first tradeoffs.

However, behavior depends on:

- Write concern
- Read concern
- Read preference
- Cluster setup

So saying “MongoDB is always CP” is oversimplified.

7. Banking Use Case

Yes, banking usually values consistency highly.

Example:

```None
Wrong balance > temporary retry error
```

So CP-style design is common.

But real banking systems often use:

- Relational DBs
- Transaction logs
- Consensus services
- Strong ACID systems

Not only MongoDB.

8. CP Behavior During Partition

If minority partition isolated:

```None
Cannot accept writes
```

If majority side alive:

```None
Continue safely
```

This prevents split-brain writes.

9. Examples of CP-Oriented Systems

- MongoDB (certain configs)
- ZooKeeper
- etcd
- HBase

10. Advantages

Strong correctness
Safer writes
Prevents conflicting updates
Good for critical data

11. Trade-offs

Temporary request failures
Higher latency sometimes
Election pauses
More coordination overhead

12. CAP Comparison

Type	Priority

CA	Correct + responsive (no partition handling)
CP	Correct + partition-safe
AP	Responsive + partition-safe

13. Interview Answer Example

Q: Why choose CP database for banking?

> Because correctness of balances and transactions is more important than always accepting requests. Temporary retry is better than inconsistent money data.

14. Short Revision Notes

```None
CP =

Consistency + Partition tolerance



During partition:

may reject requests



Used for:

banking

payments

metadata

inventory
```

15. One-Line Summary

CP databases prioritize correct and synchronized data during network failures, even if some requests must wait or fail temporarily.